

FITBUDDY AI

Phase 4: Project Planning Phase

Prepared from the current public FitBuddy_ai repository and organized according to the Phase 1–6 structure of the provided AI/ML & GenAI project template.

4.1 Project Planning Objective

The planning phase converts the design into a practical development sequence. Work is divided into environment setup, backend, database, UI, feedback workflow, AI integration and testing.

4.2 Work Breakdown Structure

Work Package	Tasks	Output
WP1 Setup	Create repository, Python environment and project structure.	Ready development environment
WP2 Backend	Configure FastAPI, lifespan and routing.	Running web backend
WP3 Database	Create SQLite schema and connection helper.	Persistent data layer
WP4 UI	Build registration, dashboard and admin templates.	Browser interface
WP5 Feedback	Capture feedback and persist updated plan.	Adaptive workflow
WP6 GenAI	Connect Gemini service, prompts, response validation and fallback handling.	GenAI-enabled generation
WP7 Testing	Unit, integration, UI and error-path tests.	Validated application
WP8 Documentation	Prepare phase documents, screenshots and demonstration material.	Submission package

4.3 Recommended Development Sequence

- 1 Confirm Python and virtual environment setup.
- 2 Install and verify FastAPI, Uvicorn, Jinja2 and configuration dependencies.
- 3 Run the current FastAPI application and verify the home page.
- 4 Verify SQLite database initialization and user table creation.
- 5 Test profile submission and dashboard rendering.
- 6 Test individual-user retrieval and admin registry.
- 7 Test feedback submission and updated-plan persistence.
- 8 Integrate Gemini through a dedicated service module.
- 9 Add prompt validation, safe output handling and fallback behavior.
- 10 Perform complete end-to-end testing and collect screenshots.

4.4 Milestones

Milestone	Completion condition
M1	Application starts successfully and home page loads.
M2	User profile is stored in SQLite.
M3	Workout plan is displayed on dashboard.
M4	Feedback produces an updated-plan record.
M5	Admin registry lists stored users.
M6	Gemini integration, if enabled, returns validated output.
M7	All planned test cases pass and documentation is complete.

4.5 Risks and Mitigation

Risk	Impact	Mitigation
Missing API key	GenAI generation unavailable.	Use .env configuration and a clear startup/configuration check.
Model/API change	Generation failure or incompatibility.	Keep model name configurable and isolate Gemini calls in one module.
Invalid form data	Bad records or server errors.	Use validation, sensible ranges and clear error messages.
Database record error	Lost or inconsistent plans.	Use parameterized SQL and test CRUD operations.
Unsafe/unsuitable generated content	Poor user experience.	Use constrained prompts, validation and human-reviewable output.

4.6 Project Completion Criteria

- All six reference phases have a corresponding deliverable.
- Core user registration, plan, feedback and admin workflows function end-to-end.
- Database records can be inspected and reproduced during demonstration.
- AI integration is either demonstrated as working or explicitly documented as pending.
- Testing evidence and screenshots are included in the final submission.

Project repository: https://github.com/asvikamoorthy-cyber/FitBuddy_ai

Reference structure: <https://github.com/Ravi-teja-777/AI-ML-and-GEN-AI-Track-Project-Template>

Scope note: This document describes the project state visible in the public repository and identifies proposed/future items separately where applicable.